1 Define machine learning.

Ans

Machine learning is a subset of AI, which enables the machine to automatically learn from data, improve performance from past experiences, and make predictions. Machine learning contains a set of algorithms that work on a huge amount of data.

2 List the types of machine learning.

Ans

- Supervised Machine Learning
- Unsupervised Machine Learning
- Semi-Supervised Machine Learning
- Reinforcement Learning

3 Explain perspectives and issues in machine learning.

- **Perspectives:** Machine learning can be viewed from different angles, including theoretical (mathematical foundations), engineering (implementation and scalability), ethical (bias and fairness), business (economic impact), and scientific (research applications).
- **Issues:** Key challenges include data quality, bias, privacy concerns, model explainability, computational cost, adversarial attacks, and job displacement. Ensuring fairness, security, and efficiency remains critical in ML development.

4 Illustrate an activation function.

Ans

An activation function is a mathematical function applied to the output of a neuron.

It helps to determine whether the neuron will fire or not. Activation Function can be considered primarily as a step function.

5 Recall the Perceptron in neural network.

Ans

The **Perceptron** is the simplest type of artificial neural network, introduced by **Frank Rosenblatt** in 1958. It is a **supervised learning algorithm** used for binary classification tasks.

A perceptron consists of **Inputs ,Weights,Bias,Activation Function**.

Mathematical Representation

The perceptron computes a weighted sum of inputs:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

where f is a step function:

$$f(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$$

6 Solve the classification and Regression Problem with an example.

Ans

- Classification Problem:
    - Example: Predicting if an email is "spam" or "not spam" (binary classification).
    - Solution: Use a Logistic Regression or Decision Tree model with labelled email data (features like word frequency).
- Regression Problem:
    - Example: Predicting house prices based on features like area, number of rooms, and location.
    - Solution: Use a Linear Regression model to find a relationship between house features and price.

7 Describe a Multi-Layer Perceptron (MLP).

Ans

A **Multi-Layer Perceptron (MLP)** is a type of artificial neural network that consists of multiple layers of neurons. It is capable of solving complex problems, including classification and regression.

Structure of MLP

1. **Input Layer** – Receives input features.
2. **Hidden Layers** -It learns the representation using activation functions.
3. **Output Layer** – Produces the final prediction

8 Compare the Different Output Activation Functions.

Ans

Function	Used For	Range	Key Point
Sigmoid	Binary Classification	(0,1)	Converts to probability but may cause vanishing gradients.
Softmax	Multi-Class Classification	(0,1) (sum = 1)	Assigns probabilities to multiple classes.
ReLU	Regression & Hidden Layers	[0, ∞)	Avoids vanishing gradients but some neurons may die.
Tanh	Binary Classification	(-1,1)	Zero-centered, better than Sigmoid in some cases.
Linear	Regression	(-∞, ∞)	Outputs any continuous value.

9 Apply The Radial Basic Function Algorithm with an example.

Ans

Example: Applying Radial Basis Function (RBF) Algorithm

Given Data:

Input: $x = [1, 2, 3, 4]$
Target: $y = [1, 4, 9, 16]$

Step 1: Choose RBF Centers

Select $c_1 = 1.5$, $c_2 = 3.5$.

Step 2: Compute Gaussian RBF

$$\phi(x) = e^{-\gamma(x-c)^2}, \quad (\text{set } \gamma = 1)$$

Step 3: Construct Matrix and Solve for Weights

Solve:

$$\begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix} = \begin{bmatrix} 0.7788 & 0.0019 \\ 0.7788 & 0.1054 \\ 0.1054 & 0.7788 \\ 0.0019 & 0.7788 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

Step 4: Predict for $x = 2.5$

$$y_{pred} = w_1 e^{-(2.5-1.5)^2} + w_2 e^{-(2.5-3.5)^2}$$

This approximates $f(2.5)$.

10 Describe The Cubic Spline.

Ans

A **cubic spline** is a smooth, piecewise **cubic polynomial** used for interpolation. It ensures **continuity** in function, first derivative, and second derivative at data points, making it useful in **curve fitting and graphics**.

11 Describe the Curse of Dimensionality.

Ans

As dimensions increase, **data becomes sparse**, computations grow, and models risk **overfitting**. This makes pattern recognition harder, requiring techniques like **PCA** to reduce dimensions.

12 Explain about Support Vector Machines.

Ans

SVM is a **supervised learning algorithm** used for **classification and regression**. It finds the **optimal hyperplane** that maximizes the margin between classes. Using **support vectors** and **kernels**, SVM efficiently handles **linear and non-linear** data.

13 Define classification and regression trees (CART).

Ans

The **CART algorithm** is a decision tree method used for **classification** (categorical output) and **regression** (continuous output).

- **Classification Trees**: Use **Gini impurity** or **entropy** to split data into classes (e.g., spam or not spam).
- **Regression Trees**: Use **Mean Squared Error (MSE)** to split data for predicting continuous values (e.g., house prices).

14 Explain the difference between boosting and bagging techniques.

Ans

- **Boosting**: Models are built **sequentially**, each correcting previous errors (e.g., AdaBoost, XGBoost). It **reduces bias** and improves weak learners.
- **Bagging**: Models are trained **independently in parallel**, combining results (e.g., Random Forest). It **reduces variance** by averaging multiple models.

15 Show how to combine multiple classifiers using voting mechanisms in ensemble learning.

Ans

In **ensemble learning**, voting combines multiple classifiers to improve accuracy.

It can implement either "hard voting" where each classifier casts a single vote for the predicted class, or "soft voting" where each classifier provides a probability distribution over classes, and the final prediction is based on the aggregated probabilities across all classifiers; the class with the highest combined probability is chosen as the final prediction.